

the time taken for return after reflection), stereo cameras (which piece together 3-dimensional information about the environment from a number of photos taken from strategically placed cameras)

- **Sensors that measure orientation and position:** tilt sensors and gyroscopes (that use the principle of conservation of angular momentum to record changes in angular orientation), magnetic direction sensors and Global Positioning System units
- **Sensors that measure electrical properties:** gaussmeters, voltage sensors, current sensors
- **Sensors that measure vibrations, chemical compositions and moisture**
- **Sensors that repair themselves, or that are a nanometer long**

What it runs on - power systems

A robot isn't going anywhere without power. Technological marvels like the Mars Rover are products of a fantastic degree of engineering skill that, among other things, has found reliable and sufficient ways to power the consumptive robotic machinery even 200 million odd kilometres from the earth. It thus becomes of prime importance to take into account the proximity to civilisation (which, in the case of Mars, is not very good), the use cases, duration of use, and type of power consumption - while designing the life's blood that powers any robotic system.



Source: Wikimedia

NASA's Helios, a solar-powered Unmanned Aerial Vehicle (UAV), theoretically capable of infinitely long flight times.